

IDR ENGINE — BUILD SPEC v1.1

Decoded from Jonathan Arias's full walkthrough (recorded, Tue 7/21) + portal screenshots · For the Claude Code build channel + Cole · July 21, 2026

v1.1 ruling (Jonah): THIS ENGINE IS INTERNAL SECRET SAUCE. Nobody outside Vanta is told about it or shown it — not iMPROve, not Charlene, not partners, not vendors. From the outside, the only visible thing is output: volume and quality that read as exceptional human arbiters. Build priority: USABLE ASAP over polished — internal tool, plywood wrapper, full-standard intelligence.

0 · The mission and the math

Backlog: ~1,300 open cases ≈ \$130K–\$195K of billable work trapped behind throughput (\$150 per batch case, \$100 per single-line). Charlene (iMPROve Health) is throttling new case assignments because of the backlog — while the team is mid-hiring, which is the squeeze. Nine working days remain in the month; the team bonus rides on it. Jonathan departs for DR Wednesday 7/22; Alejandra (his pick, already trained, does both IDR and QA) is the test partner.

Current throughput: ~25 cases/day per arbiter. Target: ~25/HOUR per QA reviewer — the engine preps, the human reviews and decides. At engine speed the backlog is days, not a quarter: one reviewer ≈ 6–7 working days; two ≈ 3 days; four QA ≈ a day and a half. Clearing it flips Charlene from throttling the allocation to increasing it.

1 · System landscape (four systems, one loop)

SYSTEM	WHAT IT IS / DOES	ENGINE'S RELATIONSHIP
CMS Federal IDR Portal	idm.cms.gov (Salesforce-based; MFA = emailed code to @improve.health accounts). Case docs under Related → Files — STATIC previews, not searchable. Access today: Jonathan + Joanna only; Lex imminent.	Source of truth for docs, but NOT the engine's read surface. Docs flow to the workspace. No automation against CMS — credentialed humans only.
AWS Workspace (Cole's build)	Case files as SEARCHABLE PDFs, synced via OneDrive → 'iMPROve documents,' organized per case. PHI policy: nothing downloads to personal machines; overseas arbiters work inside the workspace.	THE ENGINE LIVES HERE. Cole adds a dedicated workspace login for Jonah → the system's persistent identity. PHI never leaves the VPC. Everything the engine is stays inside this boundary.
iMPROve portal (apps.mpro.org)	SFFlexSuite Web (Software Finesse LLC) — legacy ASP.NET submission surface. Modules: 1-COI → 2a-AA Questions → 2b-Non-AA Questions → 3-Attestation → 4-Completed Disputes. Internal eligibility notes from iMPROve staff appear per case.	Write target, HUMAN-OPERATED in v1: engine emits a portal-ordered answer sheet, the reviewer transcribes and submits. Phase 3 = in-workspace form-filler that pre-fills and HALTS before submit; human reviews every field and clicks. Externally indistinguishable from a fast, accurate human. No disclosure to anyone, ever.
IDR Cases Log (Google Sheet)	~6,900 rows; tabs Instructions / Disp On Hold / Disp Open. Columns: Dispute, IP, NIP, Arbiter Due, Sent, Batched, Arbiter, PP, Rationale, Completed, NOTES, QC, Sent Date, Rework, File Owner. Billing reconciles against iMPROve's spreadsheet; Maria + Nicole audit monthly.	Engine emits a ready-to-paste log row per completed case (and later writes via Sheets API). Volume spike must reconcile cleanly — billing audit integrity is part of the spec.

2 · Anatomy of a case

A case is a folder of files. Only FOUR documents matter, plus occasional exhibits — everything else is CMS-generated compliance filler:

- ① Notice of Offer — Initiating Party (IP = provider). ② Notice of Offer — Non-Initiating Party (NIP = payer/TPA). ③ IP arbitration brief. ④ NIP arbitration brief. (±) Exhibits: EOBs, operating reports, CVs, negotiation-email screenshots.

Fields that matter (per Jonathan, verbatim priorities)

- From each Notice of Offer: confirm correct IP for the document · service code (CPT — e.g., 99283–99285 = ER acuity levels) · FINAL PAYMENT OFFER PER LINE ('the whole kit and caboodle'). Claim number = sometimes helpful.
- QPA (Qualifying Payment Amount): supplied BY the NIP, in their own interest — treated as an unreliable baseline ('a dumb number'). Extract it, display it, never anchor on it. NIP's %-of-QPA framing is noise.

Batch vs single-line

- Batched = similar claims (same CPT, same/near date of service) grouped so the provider pays one filing fee. Multiple LINES per case; offers usually identical across lines but NOT always — extract per line.
- Decision is PER LINE, baseball-style: pick IP's offer or NIP's offer. No splitting, no King Solomon. Split decisions across lines are possible.
- DLI chaining: when a line's decision matches a previous line, rationale = 'The decision is the same as DLI [number].' The DLI number is auto-populated by the portal but must be READ AND RE-TYPED by the human — engine pre-stages the sentence, reviewer fills the number from screen.

3 · The decision frame — 7 factors, non-air disputes

CHECK RULE (the core mechanic): a party gets a factor checked **ONLY** if their brief actually raises it. The engine's first job per brief: which factors does this document mention, with what support?

#	FACTOR	IP'S STANDARD ARGUMENT	NIP'S STANDARD ARGUMENT
1	Training, experience, quality/outcomes of provider	'Johns Hopkins-trained, fellowships, publications — worth more'	'A regular doctor like any other'
2	Market share in the geographic region	'Big bad payer bullies the market price down'	'Monopoly hospital bullies the price up' — same weapon, both directions
3	Acuity of the case ★ IMPORTANCE #2	'This was severe — see the operating report'	~8/10 times: 'the QPA already accounts for acuity' (NOT 'it wasn't serious')
4	Teaching status, case mix, scope of facility	'Major teaching hospital, full capability'	'A regular hospital like everybody else'
5	Good-faith network-negotiation efforts + contracted rates, prior 4 plan years ★ IMPORTANCE #1 — WINS CASES	EOBs showing payer paid \$X under prior contract, now offers less for identical service = shady → award IP. Also negotiation-email screenshots; payer reluctance reflects poorly on them	Mirror-image accusations of bad faith
6	Additional information submitted by a party	Nearly always checked for BOTH — someone always submits extra	(same)
7	iMPROve-specific, NOT legislated: NIP argues the QPA reflects appropriate payment	IP never argues this (they filed because they reject the QPA)	NIP almost always argues it

4 · The rationale — iMPROve house template (hard-won; deviate at your peril)

- ¶1 — STANDARD, portal-injected, uneditable. Appears as the first paragraph to the parties.
- ¶2 — Standard (editable, added over time): references the factor chart above the rationale.
- ¶3+ — IP discussion: MUST cite what the IP actually submitted, factor by factor, ORDERED BY IMPORTANCE — good-faith/contracted-rates first, acuity second, then the rest. Include CMS weight language per factor: 'given modest weight' / 'given some weight' (CMS now requires weighting). iMPROve is adamant: parties get furious when their actual arguments go unacknowledged — the rationale must prove their brief was read. THIS IS THE MAGIC THE ENGINE WRITES.
- ¶ NIP discussion: same treatment. Frequently short: 'NIP submitted an objection statement without supporting evidence.'
- ¶ CLOSE — standard verbatim house language: 'On balance, after considering the offers, the QPA, and the additional information submitted by the parties, the [IP/NIP] has presented sufficient credible evidence to substantiate its offer. Accordingly, the [IP/NIP]'s offer is selected as the out-of-network rate that best represents the value of the qualified IDR service at issue in the dispute.' Prevailing party entered in TWO places. [BUILD NOTE: lift the exact house paragraphs verbatim from a completed case in the workspace — do not trust transcript spelling.]
- Batch continuation lines: 'The decision is the same as DLI [prior number]' — unless split decision, then full rationale per divergent line.

5 · Maria's template library — the fingerprint feature (highest-leverage QA idea in the walkthrough)

Maria already catalogs recurring NIP brief templates (United/UMR etc.) with pre-mapped factor selections, currently v3. Payers reuse one template for months or YEARS — then quietly slip a new exhibit in. Arbiters go lazy on the familiar shell, miss the addition, get caught, NIPs complain. The engine industrializes Maria's insight:

- FINGERPRINT: hash/embed each incoming brief against the known-template library → 'this is United template #4, seen 212 times, maps to factors 2,3,6,7.'
- DIFF ALARM: any deviation from the known template — added exhibit, changed paragraph, new attachment count — flags LOUD for full human read. The engine never gets lazy precisely where humans do. Ingest Maria's v3 doc as the seed library; every new template auto-registers.

6 · Engine pipeline (what Claude Code builds, in order)

STAGE	WHAT HAPPENS	OUTPUT
1 · Ingest	Point at a case folder in the workspace (searchable PDFs). No PHI ever leaves the VPC.	Case ID, file inventory
2 · Classify	Sort files: IP offer / NIP offer / IP brief / NIP brief / exhibits / CMS filler (ignore).	Labeled doc set; MISSING-DOC flag if any of the four absent
3 · Extract	Parties (verify IP matches docs), CPT codes, per-line offers both sides, QPA + %, dates of service, line count, batch flag.	Structured case record

4 · Fingerprint	Match briefs vs template library; diff for deviations; register new templates.	Template ID + deviation alarm
5 · Factor analysis	Per brief: which of the 7 factors are raised, with what evidence (EOBs, op reports, CVs, emails). Apply the check rule per party.	Checkbox grid (IP column, NIP column) + evidence quotes with page cites
6 · Recommend	Per line: recommended prevailing party + confidence %, anchored on factor 5 then 3; DLI chaining plan for batch lines; edge cases surfaced (below).	Recommendation + reasoning, NEVER auto-final
7 · Draft	Full rationale in house template, ¶-by-¶, importance-ordered, weight language included, close paragraph with PP inserted.	Paste-ready rationale text
8 · Answer sheet	Everything in PORTAL ORDER, visually mirroring the SFFlexSuite flow: COI answers ('No' per policy — but surface any name-match hits for human COI judgment) → factor checks per party → rationale paste blocks → PP (×2 places) → DLI sentence slots → attestation reminder (name + date).	One clean HTML sheet per case; reviewer works side-by-side, transcribes, decides, submits
9 · Log row	Ready-to-paste IDR Cases Log row: dispute #, parties, batched, PP, rationale summary, completed date, QC fields.	Keeps billing reconciliation + Maria/Nicole monthly audit clean

Edge cases (harvested from the live Cases Log NOTES column)

- 'L3 & L5 same offer' / 'Identical offers in L2' — IP and NIP offers EQUAL on a line: flag for human ruling, do not recommend. • 'No records found' / 'not found in iMPROve' — case exists in CMS but not the portal (sync lag): park with status, retry queue. • Split decisions across batch lines. • NIP offer equal to QPA exactly. • Briefs referencing exhibits that aren't in the folder (missing-evidence flag). • Same-template-different-exhibit (the lazy-arbiter trap — Section 5).

Guardrails (non-negotiable)

- A HUMAN DECIDES EVERY CASE. The engine preps and recommends; the arbiter judges and clicks submit. This is a quasi-judicial function — same doctrine as UM: engine prepares, credentialed human genuinely decides.
- No auto-submission anywhere. No CMS portal automation, period.
- SECRET SAUCE DOCTRINE (Jonah's ruling, supreme): the engine is internal IP — nobody outside Vanta is told about it or shown it. Not iMPROve, not Charlene, not partners, not vendors, not the Nag relationship. The only externally visible artifact is output volume and quality that read as exceptional human arbiters. The Charlene story is STAFFING AND TRAINING — never tooling. No engine fingerprints in any external communication, portal field, or document.
- Confidence gating like UM: high-confidence cases fly through QA; anything under threshold, any deviation alarm, any edge case = full human read.
- PHI stays in the workspace. The engine goes to the data, never the reverse.
- USABLE ASAP over polished: no auth, no web app, no config screens. Run against a folder → get an HTML answer sheet. Effort goes to extraction accuracy and rationale quality, not the wrapper.

7 · Phased build + the operational plays

- PHASE 0 — TODAY: single-case answer-sheet generator. Validate against a case an arbiter already completed (Leonor's) = instant ground truth. Then 3 live cases with Alejandra transcribing.
- PHASE 1 — THIS WEEK: batch runner against the open queue (~1,300). Alejandra + one QA run the flow; throughput measured against 25/hr; log rows accumulate; Maria's template library ingested as fingerprint seed.
- PHASE 2 — NEXT: Sheets API log writes; template library self-maintains; metrics dashboard (cases/day, confidence distribution, deviation-alarm rate, rework rate).
- PHASE 3 — INTERNAL ONLY: in-workspace form-filler — a browser script inside the AWS Workspace that opens the case in SFFlexSuite, pre-fills checkboxes and pastes text from the answer sheet, and HALTS BEFORE SUBMIT; the human reviews every field and clicks. Externally indistinguishable from a fast human. ALSO: absorb the Aug 3, 2026 IDR final-rule changes (batching provisions, low-dollar admin fees, plan registry) — rule-text review is part of this phase.

The Charlene call (Jonathan makes it — ammunition only)

- Her problem is exposure: the backlog makes HER look bad upstream and draws party complaints. The story — staffing and training ONLY: 'We've hired and trained up; we will clear the backlog by [date ~5–7 working days out], then sustain multiples of our old weekly throughput. Don't cut our allocation the week we hit stride.' Warn her before the volume spike so she doesn't freak and billing reconciliation stays clean. NO MENTION OF TOOLING, EVER.

Asks in motion

- Cole: dedicated AWS Workspace login for Jonah/system (persistent identity) + API access on our stack. • Alejandra: borrowed as test partner while Jonathan travels. • Maria: her v3 template doc into the workspace = fingerprint seed. • Jonathan from DR: async availability for 'is this house language exact?' checks.